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CAHIER DES CHARGES

Déploiement d'une
interconnexion sécurisée VPN
IPsec site-à-site

PROJET 2

EPREUVE E6

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1. Cahier des charges – Expression des besoins

1.1. Descriptif de l'existant

L'entreprise possède deux sites géographiques distincts :

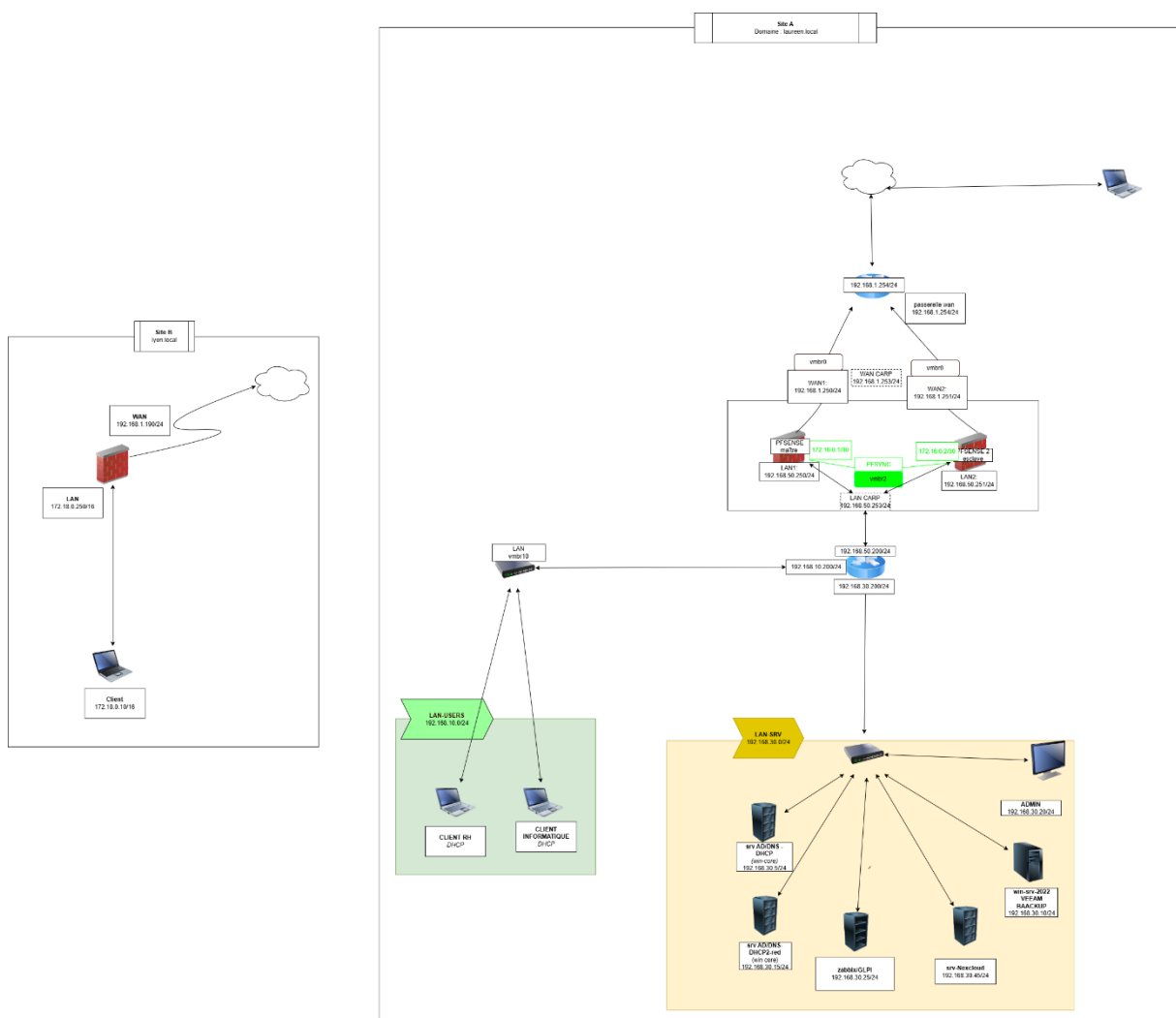
- Site principal laureen.local (Site A)
- Site distant lyon.local (Site B)

Chaque site possède :

- Un réseau local
- Un firewall
- Un accès Internet

Les réseaux ne sont pas interconnectés de manière sécurisée.

- Schéma de l'infrastructure



- [illegible]

- | Table de routage du routeur : Pfsense 1 | | | |
|---|---------------|-------------------|-------------------|
| @Réseau Destination | Masque | @Passerelle | @Interface |
| 192.168.10.0/24 | 255.255.255.0 | 192.168.50.200/24 | 192.168.50.250/24 |
| 192.168.30.0/24 | 255.255.255.0 | 192.168.50.200/24 | 192.168.50.250/24 |
| 192.168.50.0/24 | 255.255.255.0 | 192.168.50.250/24 | 192.168.50.250/24 |
| 0.0.0.0 | 0.0.0.0 | 192.168.1.254/24 | 192.168.1.250/24 |
| | | | |
| Table de routage du routeur : Pfsense 2 | | | |
| @Réseau Destination | Masque | @Passerelle | @Interface |
| 192.168.10.0/24 | 255.255.255.0 | 192.168.50.200/24 | 192.168.50.251/24 |
| 192.168.30.0/24 | 255.255.255.0 | 192.168.50.200/24 | 192.168.50.251/24 |
| 192.168.50.0/24 | 255.255.255.0 | 192.168.50.251/24 | 192.168.50.251/24 |
| 0.0.0.0 | 0.0.0.0 | 192.168.1.254/24 | 192.168.1.251/24 |
| | | | |
| Table de routage du routeur : Routeur | | | |
| @Réseau Destination | Masque | @Passerelle | @Interface |
| 192.168.30.0/24 | 255.255.255.0 | 192.168.30.200/24 | 192.168.30.200/24 |
| 192.168.10.0/24 | 255.255.255.0 | 192.168.10.200/24 | 192.168.10.200/24 |
| 192.168.50.0/24 | 255.255.255.0 | 192.168.50.200/24 | 192.168.50.200/24 |
| 172.18.0.0/16 | 255.255.0.0 | 192.168.50.253/24 | 192.168.50.200/24 |
| 0.0.0.0 | 0.0.0.0 | 192.168.50.253/24 | 192.168.50.200/24 |
| | | | |

L'entreprise souhaite pouvoir accéder au site de Lyon à partir du site de Grenoble afin d'accéder aux fichiers.

1.3. Contraintes

1.3.1. Temps

- 1 jour maximum

1.3.2. Budget

- Budget limité : IPsec natif

1.3.3. Juridique

- Confidentialité des données et RGPD
- Utilisation du chiffrement IPsec standard

1.3.4. Organisation

- Intervention réalisée par l'administrateur système.

2. Analyse

2.1. Descriptifs des solutions

Solutions possibles :

- VPN IPsec
- OpenVPN
- WireGuard

2.2. Comparaison des solutions

Solutions	Avantages	Inconvénients
IPsec	Standard, sécurisé	Configuration complexe
OpenVPN	Facile	Moins intégré
Wireguard	Rapide	Moins sécurisé

2.3. Choix d'une solution – Argumentation

Le VPN IPsec natif est retenu car il est très utilisé par sa garantie de sécurité et ne tient en compte aucun frais supplémentaire. De plus, il est compatible avec tous les équipements.

2.4. Etude de l'impact sur le SI existant

2.4.1. Sécurité

Amélioration de la sécurité :

- Chiffrement des données
- Authentification des sites

2.4.2. Performance

Impact faible

2.4.3. Juridique

Protection des données conforme RGPD

2.4.4. Ergonomie

Transparent pour les utilisateurs

2.5. Prévision des tests de validation, à quel moment seront effectués les tests.

Test de connectivité entre les sites.

Voir le rapport de test à la fin de l'annexe jointe.

2.6. Déploiement

2.6.1. Procédure

Etape 1 : Configuration des machines du site B

Etape 2 : Mise en place du VPN Ipsec – Pfsense A

Etape 3 : Mise en place du VPN IPsec sur pfsense B

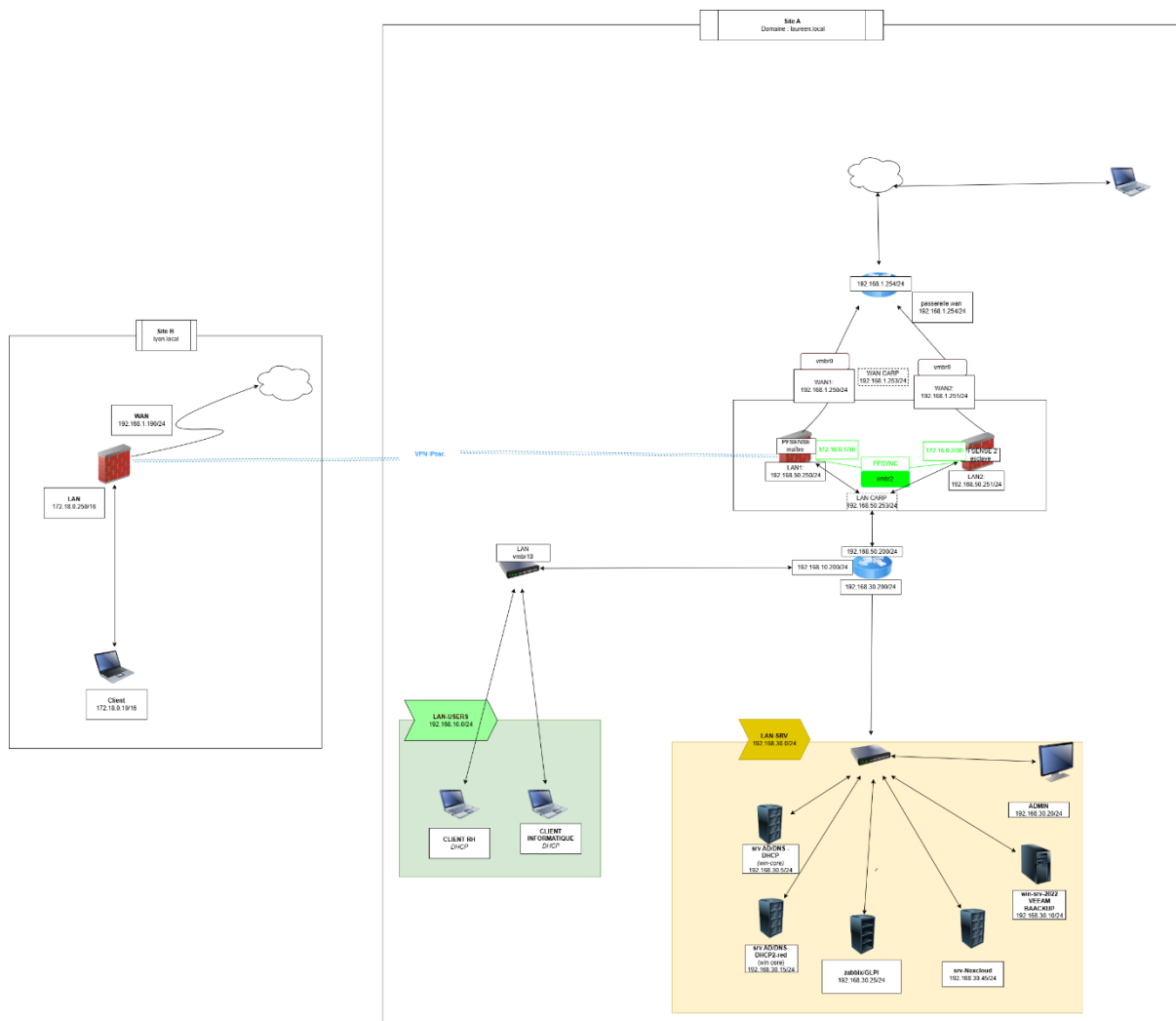
Etape 4 : Règles de filtrage

Etape 5 : Test

3.Mise en place

3.1. Descriptif de ce qui a été mis en place

- Schéma infrastructure



- Table de routage

Table de routage du routeur : Pfsense 1			
@Réseau Destination	Masque	@Passerelle	@Interface
192.168.10.0/24	255.255.255.0	192.168.50.200/24	192.168.50.250/24
192.168.30.0/24	255.255.255.0	192.168.50.200/24	192.168.50.250/24
192.168.50.0/24	255.255.255.0	192.168.50.250/24	192.168.50.250/24
0.0.0.0	0.0.0.0	192.168.1.254/24	192.168.1.250/24
Table de routage du routeur : Pfsense 2			
@Réseau Destination	Masque	@Passerelle	@Interface
192.168.10.0/24	255.255.255.0	192.168.50.200/24	192.168.50.251/24
192.168.30.0/24	255.255.255.0	192.168.50.200/24	192.168.50.251/24
192.168.50.0/24	255.255.255.0	192.168.50.251/24	192.168.50.251/24
0.0.0.0	0.0.0.0	192.168.1.254/24	192.168.1.251/24
Table de routage du routeur : Routeur			
@Réseau Destination	Masque	@Passerelle	@Interface
192.168.30.0/24	255.255.255.0	192.168.30.200/24	192.168.30.200/24
192.168.10.0/24	255.255.255.0	192.168.10.200/24	192.168.10.200/24
192.168.50.0/24	255.255.255.0	192.168.50.200/24	192.168.50.200/24
172.18.0.0/16	255.255.0.0	192.168.50.253/24	192.168.50.200/24
0.0.0.0	0.0.0.0	192.168.50.253/24	192.168.50.200/24

3.2. Méthodologie

- Configuration du tunnel IPsec – site A
- Configuration du tunnel IPsec – site B
- Test

3.3. Les procédures spécifiques seront documentées dans des liens externes à ce document

Voir annexe en fin de document

3.4. Configuration

3.5. Réalisation des tests

- Connexion au site distant : OK

Tests de connexion entre les deux sites réalisés à la fin du déploiement

3.6. Rapports de tests et de déploiement

Voir annexe à la fin de la documentation.

4. Bilan

Le site « laureen.local » possède actuellement un accès à distance sécurisé avec le site B : lyon.local. La mission a été réalisée avec succès et dans les temps.

Annexe – Procédure : Mise en place d'un VPN IPsec site-à-site

Etape 1 : Configuration des machines du site B

Pfsense site B

```
The IPv4 LAN address has been set to 172.18.0.250/16
You can now access the webConfigurator by opening the following URL in your web browser:

      https://172.18.0.250/

Press <ENTER> to continue.
QEMU Guest - Netgate Device ID: 76acb393f519ff323f5e

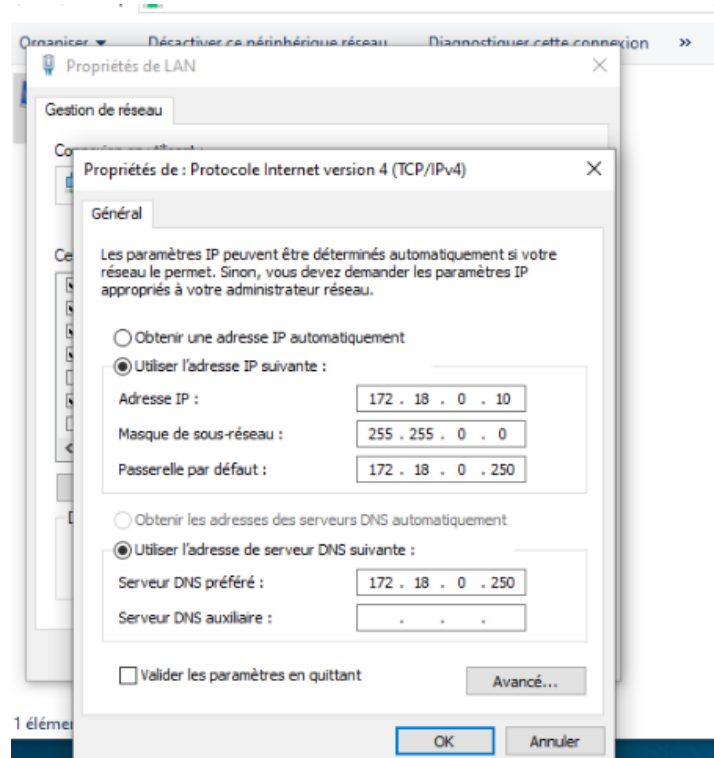
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> vtnet0      -> v4: 192.168.1.190/24
LAN (lan)      -> vtnet1      -> v4: 172.18.0.250/16

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: 
```

Client site B



Etape 2 : Mise en place du VPN Ipsec – Pfsense A

1. Configuration de la phase 1 - Pfsense A

The first screenshot shows the Pfsense Community Edition web interface. The 'VPN' menu is expanded, showing options for IPsec, L2TP, and OpenVPN. A warning message is visible at the top: 'WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.'

The second screenshot shows the 'VPN / IPsec / Tunnels' page. It features a table with columns for ID, IKE, Remote Gateway, Auth/Mode, P1 Protocol, P1 Transforms, P1 DH-Group, P1 Description, and Actions. A green '+ Add P1' button is located at the bottom right of the table.

The third screenshot shows the 'Edit Phase 1' configuration page. It is divided into two main sections: 'General Information' and 'IKE Endpoint Configuration'.
General Information:
- Description: A text input field with a placeholder: 'A description may be entered here for administrative reference (not parsed).'
- Disabled: A checkbox with the label 'Set this option to disable this phase1 without removing it from the list.'
- IKE ID: A text input field containing the value '1'.
IKE Endpoint Configuration:
- Key Exchange version: A dropdown menu set to 'IKEv2'. Below it, a note says: 'Select the Internet Key Exchange protocol version to be used. Auto uses IKEv2 when initiator, and accepts either IKEv1 or IKEv2 as responder.'
- Internet Protocol: A dropdown menu set to 'IPv4'. Below it, a note says: 'Select the Internet Protocol family.'
- Interface: A dropdown menu set to 'WAN'. Below it, a note says: 'Select the interface for the local endpoint of this phase1 entry.'
- Remote Gateway: A text input field containing the value '192.168.1.190'.

Ici, il s'agit de l'adresse WAN du deuxième Pare-feu avec lequel il va communiquer.

Phase 1 Proposal (Authentication)				
Authentication Method	Mutual PSK ▼			
	Must match the setting chosen on the remote side.			
My identifier	My IP address ▼			
Peer identifier	Peer IP address ▼			
Pre-Shared Key	pfsense123			
	Enter the Pre-Shared Key string. This key must match on both peers. This key should be long and random to protect the tunnel and its contents. A weak Pre-Shared Key can lead to a tunnel compromise.			
	 Generate new Pre-Shared Key			

Phase 1 Proposal (Encryption Algorithm)				
Encryption Algorithm	AES ▼	256 bits ▼	SHA256 ▼	14 (2048 bit) ▼  Delete
	AES256-GCM ▼	128 bits ▼	SHA256 ▼	14 (2048 bit) ▼  Delete
	Algorithm	Key length	Hash	DH Group
Note: SHA1 and DH groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.				
Add Algorithm	 Add Algorithm			

Expiration and Replacement	
Life Time	28800 ▼ Hard IKE SA life time, in seconds, after which the IKE SA will be expired. Must be larger than Rekey Time and Reauth Time. Cannot be set to the same value as Rekey Time or Reauth Time. If left empty, defaults to 110% of whichever timer is higher (reauth or rekey)
Rekey Time	25920 ▼ Time, in seconds, before an IKE SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Only supported by IKEv2, and is recommended for use with IKEv2. Leave blank to use a default value of 90% Life Time when using IKEv2. Enter a value of 0 to disable.
Reauth Time	0 ▼ Time, in seconds, before an IKE SA is torn down and recreated from scratch, including authentication. This can be disruptive unless both sides support make-before-break and overlapping IKE SA entries. Cannot be set to the same value as Life Time. Supported by IKEv1 and IKEv2. Leave blank to use a default value of 90% Life Time when using IKEv1. Enter a value of 0 to disable.
Rand Time	2880 ▼ A random value up to this amount will be subtracted from Rekey Time/Reauth Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Advanced Options

Child SA Start Action

Default

Set this option to force specific initiation/responder behavior for child SA (P2) entries

Child SA Close Action

Default

Set this option to control the behavior when the remote peer unexpectedly closes a child SA (P2)

NAT Traversal

Auto

Set this option to enable the use of NAT-T (i.e. the encapsulation of ESP in UDP packets) if needed, which can help with clients that are behind restrictive firewalls.

MOBIKE

Disable

Set this option to control the use of MOBIKE

Gateway duplicates

☐ Enable this to allow multiple phase 1 configurations with the same endpoint. When enabled, pfSense does not manage routing to the remote gateway and traffic will follow the default route without regard for the chosen interface. Static routes can override this behavior.

Split connections

☐ Enable this to split connection entries with multiple phase 2 configurations. Required for remote endpoints that support only a single traffic selector per child SA.

PRF Selection

☐ Enable manual Pseudo-Random Function (PRF) selection
Manual PRF selection is typically not required, but can be useful in combination with AEAD Encryption Algorithms such as AES-GCM

Custom IKE/NAT-T Ports

Remote IKE Port

Remote NAT-T Port

UDP port for IKE on the remote gateway. Leave empty for default automatic behavior (500/4500).

UDP port for NAT-T on the remote gateway.

Dead Peer Detection

☒ Enable DPD
Check the liveness of a peer by using IKEv2 INFORMATIONAL exchanges or IKEv1 R_U_THERE messages. Active DPD checking is only enforced if no IKE or ESP/AH packet has been received for the configured DPD delay.

Delay

10

Delay between sending peer acknowledgement messages. In IKEv2, a value of 0 sends no additional messages and only standard messages (such as those to rekey) are used to detect dead peers.

Max failures

5

Number of consecutive failures allowed before disconnecting. This only applies to IKEv1; in IKEv2 the [retransmission timeout](#) is used instead.

Save

VPN / IPsec / Tunnels

Tunnels

Mobile Clients

Pre-Shared Keys

Advanced Settings

IPsec Tunnels

	ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐	Disable	1	V2	WAN	Mutual PSK	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)	

2. Configuration de la Phase 2 – Pfsense A

Nous ajoutons les 3 réseaux du site A : laureen.local

- 1^{ère} phase de la phase 2

VPN / IPsec / Tunnels / Edit Phase 2

Tunnels

Mobile Clients

Pre-Shared Keys

Advanced Settings

General Information

Description

A description may be entered here for administrative reference (not parsed).

Disabled

☐ Disable this phase 2 entry without removing it from the list.

Mode

Tunnel IPv4

Phase 1

No description (IKE ID 1)

P2 reqid

1

Networks

Local Network

LAN subnet

Type

Address

Local network component of this IPsec security association.

NAT/BINAT translation

None

Type

Address

If NAT/BINAT is required on this network specify the address to be translated

Remote Network

Network

Type

Address

Remote network component of this IPsec security association.

Phase 2 Proposal (SA/Key Exchange)

Protocol ESP
 Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.

Encryption Algorithms ☒ AES 256 bits

☐ AES128-GCM Auto

☐ AES192-GCM Auto

☒ AES256-GCM 128 bits

☐ CHACHA20-POLY1305

Hash Algorithms ☐ SHA1 ☒ SHA256 ☐ SHA384 ☐ SHA512 ☐ AES-XCBC
 Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.

PFS key group 14 (2048 bit)
 Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time 3600
 Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3960.

Rekey Time 3240
 Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.

Rand Time 360
 A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Keep Alive

Automatically ping host 172.18.0.250
 Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.

Keep Alive ☐ Enable periodic keep alive check
 Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.



VPN / IPsec / Tunnels

Tunnels Mobile Clients Pre-Shared Keys Advanced Settings

IPsec Tunnels

ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐ Disable 1	V2	WAN 192.168.1.190	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)		  

ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐ Disable 1	tunnel	LAN	172.18.0.0/16	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		  

- 2^{ème} phase de la phase 2

VPN / IPsec / Tunnels / Edit Phase 2

Tunnels Mobile Clients Pre-Shared Keys Advanced Settings

General Information

Description
A description may be entered here for administrative reference (not parsed).

Disabled ☐ Disable this phase 2 entry without removing it from the list.

Mode Tunnel IPv4

Phase 1 No description (IKE ID 1) 

P2 reqid 2

Networks

Local Network Network 192.168.30.0 / 24
Type Address
Local network component of this IPsec security association.

NAT/BINAT translation None / 0
Type Address
If NAT/BINAT is required on this network specify the address to be translated

Remote Network Network 172.18.0.0 / 16
Type Address

Phase 2 Proposal (SA/Key Exchange)	
Protocol	ESP Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.
Encryption Algorithms	☒ AES 256 bits ☐ AES128-GCM Auto ☐ AES192-GCM Auto ☒ AES256-GCM 128 bits ☐ CHACHA20-POLY1305
Hash Algorithms	☐ SHA1 ☒ SHA256 ☐ SHA384 ☐ SHA512 ☐ AES-XCBC Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.
PFS key group	14 (2048 bit) Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement	
Life Time	3600 Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3960.
Rekey Time	3240 Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.
Rand Time	360 A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.
Keep Alive	
Automatically ping host	Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.
Keep Alive	☐ Enable periodic keep alive check Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.
Save	

VPN / IPsec / Tunnels

Tunnels

Mobile Clients

Pre-Shared Keys

Advanced Settings

IPsec Tunnels

	ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐	1	V2	WAN 192.168.1.190	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)		

	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐	1	tunnel	LAN	172.18.0.0/16	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		
☐	2	tunnel	192.168.30.0/24	172.18.0.0/16	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		

- 3^{ème} phase de la phase 2

VPN / IPsec / Tunnels / Edit Phase 2

Tunnels

Mobile Clients

Pre-Shared Keys

Advanced Settings

General Information

Description

A description may be entered here for administrative reference (not parsed).

Disabled

☐ Disable this phase 2 entry without removing it from the list.

Mode

Tunnel IPv4

Phase 1

No description (IKE ID 1)

P2 reqid

3

Networks

Local Network

Network

192.168.10.0 / 24

Type Address

Local network component of this IPsec security association.

NAT/BINAT translation

None

/ 0

Type Address

If NAT/BINAT is required on this network specify the address to be translated

Remote Network

Network

172.18.0.0 / 16

Type Address

Phase 2 Proposal (SA/Key Exchange)

Protocol

ESP

Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.

Encryption Algorithms

☒ AES

256 bits

☐ AES128-GCM

Auto

☐ AES192-GCM

Auto

☒ AES256-GCM

128 bits

☐ CHACHA20-POLY1305

Hash Algorithms

☐ SHA1

☒ SHA256

☐ SHA384

☐ SHA512

☐ AES-XCBC

Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.

PFS key group

14 (2048 bit)

Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time

3600

Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3960.

Rekey Time

3240

Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.

Rand Time

360

A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Keep Alive

Automatically ping host

Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.

Keep Alive

☐ Enable periodic keep alive check

Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.

Save

VPN / IPsec / Tunnels

Tunnels Mobile Clients Pre-Shared Keys Advanced Settings

IPsec Tunnels

ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐ Disable 1	V2	WAN 192.168.1.190	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)		

ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐ Disable 1	tunnel	LAN	172.18.0.0/16	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		
☐ Disable 2	tunnel	192.168.30.0/24	172.18.0.0/16	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		
☐ Disable 3	tunnel	192.168.10.0/24	172.18.0.0/16	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		

[+ Add P2](#)

Activier Windows

[+ Add P1](#) [Delete P1s](#)

Etape 3 : Mise en place du VPN IPsec sur pfSense B

1. Configuration de la phase 1

pfSense COMMUNITY EDITION

System Interfaces Firewall Services **VPN** Status Diagnostics Help

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Firewall / Rules / LAN1

The changes have been applied successfully. The firewall rules are now reloading in the background.

VPN / IPsec / Tunnels

Tunnels Mobile Clients Pre-Shared Keys Advanced Settings

IPsec Tunnels

ID	IKE	Remote Gateway	Auth/Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
----	-----	----------------	-----------	-------------	---------------	-------------	----------------	---------

[+ Add P1](#)

VPN / IPsec / Tunnels / Edit Phase 1

Tunnels
Mobile Clients
Pre-Shared Keys
Advanced Settings

General Information

Description

A description may be entered here for administrative reference (not parsed).

Disabled

☐

Set this option to disable this phase1 without removing it from the list.

IKE ID

1

IKE Endpoint Configuration

Key Exchange version

IKEv2

Select the Internet Key Exchange protocol version to be used. Auto uses IKEv2 when initiator, and accepts either IKEv1 or IKEv2 as responder.

Internet Protocol

IPv4

Select the Internet Protocol family.

Interface

WAN

Select the interface for the local endpoint of this phase1 entry.

Remote Gateway

192.168.1.250

Activer Windows

Accédez aux paramètres pour activer Windows.

Il s'agit de l'adresse WAN du Pfsense A avec lequel il va communiquer.

Phase 1 Proposal (Authentication)

Authentication Method

Mutual PSK

Must match the setting chosen on the remote side.

My identifier

My IP address

Peer identifier

Peer IP address

Pre-Shared Key

pfsense123

Enter the Pre-Shared Key string. This key must match on both peers.
This key should be long and random to protect the tunnel and its contents. A weak Pre-Shared Key can lead to a tunnel compromise.

Generate new Pre-Shared Key

Phase 1 Proposal (Encryption Algorithm)				
Encryption Algorithm	AES	256 bits	SHA256	14 (2048 bit) 
	AES256-GCM	128 bits	SHA256	14 (2048 bit) 
	Algorithm	Key length	Hash	DH Group
Note: SHA1 and DH groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.				
Add Algorithm	 Add Algorithm			

Expiration and Replacement	
Life Time	28800 Hard IKE SA life time, in seconds, after which the IKE SA will be expired. Must be larger than Rekey Time and Reauth Time. Cannot be set to the same value as Rekey Time or Reauth Time. If left empty, defaults to 110% of whichever timer is higher (reauth or rekey)
Rekey Time	25920 Time, in seconds, before an IKE SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Only supported by IKEv2, and is recommended for use with IKEv2. Leave blank to use a default value of 90% Life Time when using IKEv2. Enter a value of 0 to disable.
Reauth Time	0 Time, in seconds, before an IKE SA is torn down and recreated from scratch, including authentication. This can be disruptive unless both sides support make-before-break and overlapping IKE SA entries. Cannot be set to the same value as Life Time. Supported by IKEv1 and IKEv2. Leave blank to use a default value of 90% Life Time when using IKEv1. Enter a value of 0 to disable.

Advanced Options	
Child SA Start Action	Default Set this option to force specific initiation/responder behavior for child SA (P2) entries
Child SA Close Action	Default Set this option to control the behavior when the remote peer unexpectedly closes a child SA (P2)
NAT Traversal	Auto Set this option to enable the use of NAT-T (i.e. the encapsulation of ESP in UDP packets) if needed, which can help with clients that are behind restrictive firewalls.
MOBIKE	Disable Set this option to control the use of MOBIKE
Gateway duplicates	☐ Enable this to allow multiple phase 1 configurations with the same endpoint. When enabled, pfSense does not manage routing to the remote gateway and traffic will follow the default route without regard for the chosen interface. Static routes can override this behavior.
Split connections	☐ Enable this to split connection entries with multiple phase 2 configurations. Required for remote endpoints that support only a single traffic selector per child SA.
PRF Selection	☐ Enable manual Pseudo-Random Function (PRF) selection Manual PRF selection is typically not required, but can be useful in combination with AEAD Encryption Algorithms such as AES-GCM
Custom IKE/NAT-T Ports	Remote IKE Port UDP port for IKE on the remote gateway. Leave empty for default automatic behavior (500/4500). Remote NAT-T Port UDP port for NAT-T on the remote gateway. 

Dead Peer Detection ☒ Enable DPD

Check the liveness of a peer by using IKEv2 INFORMATIONAL exchanges or IKEv1 R_U_THERE messages. Active DPD checking is only enforced if no IKE or ESP/AH packet has been received for the configured DPD delay.

Delay

Delay between sending peer acknowledgement messages. In IKEv2, a value of 0 sends no additional messages and only standard messages (such as those to rekey) are used to detect dead peers.

Max failures

Number of consecutive failures allowed before disconnecting. This only applies to IKEv1; in IKEv2 the [retransmission timeout](#) is used instead.

 Save

VPN / IPsec / Tunnels    

[Tunnels](#) [Mobile Clients](#) [Pre-Shared Keys](#) [Advanced Settings](#)

IPsec Tunnels									
	ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
		1	V2	WAN 192.168.1.250	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)	  

 Show Phase 2 Entries (3)

2. Configuration des phases 2

Nous ajoutons les 3 réseaux du site A : laureen.local

- 1^{ère} phase de la phase 2

Tunnels Mobile Clients Pre-Shared Keys Advanced Settings

General Information

Description

A description may be entered here for administrative reference (not parsed).

Disabled

☐ Disable this phase 2 entry without removing it from the list.

Mode

Tunnel IPv4

Phase 1

No description (IKE ID 1)

P2 reqid

1

Networks

Local Network

LAN subnet

Type

Address

Local network component of this IPsec security association.

NAT/BINAT translation

None

Type

Address

If NAT/BINAT is required on this network specify the address to be translated

Remote Network

Network

192.168.50.0

24

Address

Phase 2 Proposal (SA/Key Exchange)

Protocol

ESP

Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.

Encryption Algorithms

☒ AES

256 bits

☐ AES128-GCM

Auto

☐ AES192-GCM

Auto

☒ AES256-GCM

128 bits

☐ CHACHA20-POLY1305

Hash Algorithms

☐ SHA1

☒ SHA256

☐ SHA384

☐ SHA512

☐ AES-XCBC

Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.

PFS key group

14 (2048 bit)

Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time

Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3960.

Rekey Time

Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.

Rand Time

A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Keep Alive

Automatically ping host

Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.

Keep Alive

☐ Enable periodic keep alive check

Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.

 Save

VPN / IPsec / Tunnels



Tunnels Mobile Clients Pre-Shared Keys Advanced Settings

The changes have been applied successfully.

IPsec Tunnels

	ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐ 	Disable	1	V2	WAN 192.168.1.250	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)	  
	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐ 	Disable	1	tunnel	LAN	192.168.50.0/24	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256	  

- 2^{ème} phase de la phase 2

Tunnels
Mobile Clients
Pre-Shared Keys
Advanced Settings

General Information

Description

A description may be entered here for administrative reference (not parsed).

Disabled
☐ Disable this phase 2 entry without removing it from the list.

Mode
Tunnel IPv4

Phase 1
No description (IKE ID 1)

P2 reqid
2

Networks

Local Network
LAN subnet
 / 0

Type Address

Local network component of this IPsec security association.

NAT/BINAT translation
None
 / 0

Type Address

If NAT/BINAT is required on this network specify the address to be translated

Remote Network
Network
 192.168.30.0 / 24

Type Address

Phase 2 Proposal (SA/Key Exchange)

Protocol
ESP

Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.

Encryption Algorithms
☒ AES
 256 bits

☐ AES128-GCM
 Auto

☐ AES192-GCM
 Auto

☒ AES256-GCM
 128 bits

☐ CHACHA20-POLY1305

Hash Algorithms
☐ SHA1
☒ SHA256
☐ SHA384
☐ SHA512
☐ AES-XCBC

Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.

PFS key group
 14 (2048 bit)

Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time

3600

Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3960.

Rekey Time

3240

Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.

Rand Time

360

A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Keep Alive

Automatically ping host

Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.

Keep Alive

☐ Enable periodic keep alive check

Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.

Save

Activer Windows

VPN / IPsec / Tunnels

Tunnels

Mobile Clients

Pre-Shared Keys

Advanced Settings

The changes have been applied successfully.

IPsec Tunnels

	ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐ Disable	1	V2	WAN 192.168.1.250	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)		

	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐ Disable	1	tunnel	LAN	192.168.50.0/24	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		
☐ Disable	2	tunnel	LAN	192.168.30.0/24	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		

- 3^{ème} phase de la phase 2

General Information	
Description	 A description may be entered here for administrative reference (not parsed).
Disabled	☐ Disable this phase 2 entry without removing it from the list.
Mode	Tunnel IPv4
Phase 1	No description (IKE ID 1)
P2 reqid	3

Networks	
Local Network	LAN subnet Type Address Local network component of this IPsec security association.
NAT/BINAT translation	None Type Address If NAT/BINAT is required on this network specify the address to be translated
Remote Network	Network Type 192.168.10.0 Address Remote network component of this IPsec security association.

Phase 2 Proposal (SA/Key Exchange)	
Protocol	ESP Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.
Encryption Algorithms	☒ AES 256 bits ☐ AES128-GCM Auto ☐ AES192-GCM Auto ☒ AES256-GCM 128 bits ☐ CHACHA20-POLY1305
Hash Algorithms	☐ SHA1 ☒ SHA256 ☐ SHA384 ☐ SHA512 ☐ AES-XCBC Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.
PFS key group	14 (2048 bit) Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Expiration and Replacement

Life Time

3600

Hard Child SA life time, in seconds, after which the Child SA will be expired. Must be larger than Rekey Time. Cannot be set to the same value as Rekey Time. If left empty, defaults to 110% of Rekey Time. If both Life Time and Rekey Time are empty, defaults to 3960.

Rekey Time

3240

Time, in seconds, before a Child SA establishes new keys. This works without interruption. Cannot be set to the same value as Life Time. Leave blank to use a default value of 90% Life Time. If both Life Time and Rekey Time are empty, defaults to 3600. Enter a value of 0 to disable, but be aware that when rekey is disabled, connections can be interrupted while new Child SA entries are negotiated.

Rand Time

360

A random value up to this amount will be subtracted from Rekey Time to avoid simultaneous renegotiation. If left empty, defaults to 10% of Life Time. Enter 0 to disable randomness, but be aware that simultaneous renegotiation can lead to duplicate security associations.

Keep Alive

Automatically ping host

Sends an ICMP echo request inside the tunnel to the specified IP Address. Can trigger initiation of a tunnel mode P2, but does not trigger initiation of a VTI mode P2.

Keep Alive

☐ Enable periodic keep alive check

Periodically check this P2 and initiate it if disconnected; does not send traffic inside the tunnel. This check ignores the P1 option "Child SA Start Action" and works for both VTI and tunnel mode P2s. For IKEv2 without split connections, this only needs to be enabled on one P2.

Tunnels

Mobile Clients

Pre-Shared Keys

Advanced Settings

The changes have been applied successfully.

IPsec Tunnels

	ID	IKE	Remote Gateway	Auth/ Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐ Disable	1	V2	WAN 192.168.1.250	Mutual PSK -	AES (256 bits) AES256-GCM (128 bits)	SHA256 SHA256	14 (2048 bit) 14 (2048 bit)		

	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐ Disable	1	tunnel	LAN	192.168.50.0/24	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		
☐ Disable	2	tunnel	LAN	192.168.30.0/24	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		
☐ Disable	3	tunnel	LAN	192.168.10.0/24	ESP	AES (256 bits), AES256-GCM (128 bits)	SHA256		

Add P2

Etape 4 : Règles de filtrage

- **Pfsense A**

- LAN

VPN IPsec											
☐		0/0 B	IPv4 TCP/UDP	LAN subnets	*	172.18.0.0/16	*	*	none	LAN vers LAN du site B	
☐		0/0 B	IPv4 ICMP any	LAN subnets	*	172.18.0.0/16	*	*	none	Ping LAN vers LAN site B	

- IPsec

Firewall / Rules / IPsec

Floating

WAN

LAN

PFSYNC

IPsec

OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	✓	0/68 KiB	IPv4 TCP	*	*	*	*	none			
	✓	0/2 KiB	IPv4 ICMP any	172.18.0.0/16	*	LAN_ALL	*	*	none		
	✓	0/4 KiB	IPv4 *	172.18.0.0/16	*	LAN_ALL	*	*	none	Réseau interne accessible	

Add

Add

Delete

Toggle

Copy

Save

Separator

- **Pfsense B**

- WAN

Ces règles permettent d'autoriser le tunnel IPsec (IKE et NAT-T) sur l'interface WAN.

- LAN

VPN IPsec											
☐		0/0 B	IPv4 *	LAN subnets	*	LAN_laureen	*	*	none	LAN vers réseau laureen.local	
☐		0/0 B	IPv4 ICMP any	LAN subnets	*	LAN_laureen	*	*	none	PING du site B vers LAN du site B LAN SRV	

➤ IPsec

Firewall / Rules / IPsec

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor the filter reload progress.](#)

Floating WAN LAN **IPsec**

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 *	LAN subnets	*	192.168.30.0/24	*	*	none		
☐	✓	0/0 B	IPv4 ICMP any	LAN subnets	*	192.168.30.0/24	*	*	none	ICMP LAN du site B vers LAN du site A LAN SRV	
☐	✗	0/0 B	IPv4 *	*	*	*	*	*	none		

Add
 Add
 Delete
 Toggle
 Copy
 Save
 Separator

Etape 5 : Test

Depuis le client-admin du site A : laureen.local

Status / IPsec / Overview

Overview Leases SADs SPDs

IPsec Status

ID	Description	Local	Remote	Role	Timers	Algo	Status
con1 #12		ID: 192.168.1.250 Host: 192.168.1.250:500 SPI: 82d4a1ff5e68e82e	ID: 192.168.1.190 Host: 192.168.1.190:500 SPI: 579d41932f7b9632	IKEv2 Initiator	Rekey: 25190s (06:59:50) Reauth: Disabled	AES_CBC (256) HMAC_SHA2_256_128 PRF_HMAC_SHA2_256 MODP_2048	Established 278 seconds (00:04:38) ago

ID	Description	Local	SPI(s)	Remote	Times	Algo	Stats
con1: #5	Multiple	192.168.10.0/24 192.168.30.0/24 192.168.50.0/24	Local: c9818f6a Remote: cec311c5	172.18.0.0/16	Rekey: 2862s (00:47:42) Life: 3322s (00:55:22) Install: 278s (00:04:38)	AES_CBC (256) HMAC_SHA2_256_128 IPComp: None	Bytes-In: 1,404 (1 KiB) Packets-In: 19 Bytes-Out: 2,708 (3 KiB) Packets-Out: 19 Installed

```

Microsoft Windows [version 10.0.19045.6456]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\svc_admin>ping 172.18.0.250

Envoi d'une requête 'Ping' 172.18.0.250 avec 32 octets de données :
Réponse de 172.18.0.250 : octets=32 temps=3 ms TTL=62
Réponse de 172.18.0.250 : octets=32 temps=3 ms TTL=62
Réponse de 172.18.0.250 : octets=32 temps=2 ms TTL=62
Réponse de 172.18.0.250 : octets=32 temps=2 ms TTL=62

Statistiques Ping pour 172.18.0.250:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 2ms, Maximum = 3ms, Moyenne = 2ms

C:\Users\svc_admin>ping 172.18.0.10

Envoi d'une requête 'Ping' 172.18.0.10 avec 32 octets de données :
Réponse de 172.18.0.10 : octets=32 temps=3 ms TTL=125
Réponse de 172.18.0.10 : octets=32 temps=3 ms TTL=125
Réponse de 172.18.0.10 : octets=32 temps=3 ms TTL=125
Réponse de 172.18.0.10 : octets=32 temps=3 ms TTL=125

Statistiques Ping pour 172.18.0.10:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 3ms, Maximum = 3ms, Moyenne = 3ms

```

Depuis le client du site B : lyon.local

Status / IPsec / Overview

Overview Leases SADs SPDs

ID	Description	Local	Remote	Role	Timers	Algo	Status
con1 #3		ID: 192.168.1.190 Host: 192.168.1.190:500 SPI: 579d41932f7b9632	ID: 192.168.1.250 Host: 192.168.1.250:500 SPI: 82d4a1ff5e68e82e	IKEv2 Responder	Rekey: 24613s (06:50:13) Reauth: Disabled	AES_CBC (256) HMAC_SHA2_256_128 PRF_HMAC_SHA2_256 MODP_2048	Established 360 seconds (00:06:00) ago Disconnect P1

ID	Description	Local	SPI(s)	Remote	Times	Algo	Stats
con1: #5	Multiple	172.18.0.0/16	Local: cec311c5 Remote: c9818f6a	192.168.10.0/24 192.168.30.0/24 192.168.50.0/24	Rekey: 2582s (00:43:02) Life: 3240s (00:54:00) Install: 360s (00:06:00)	AES_CBC (256) HMAC_SHA2_256_128 IPComp: None	Bytes-In: 1,764 (2 KiB) Packets-In: 25 Bytes-Out: 3,452 (3 KiB) Packets- Out: 25 Installed Disconnect P2

```
Microsoft Windows [version 10.0.19045.6466]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\Admi>ping 192.168.50.250

Envoi d'une requête 'Ping' 192.168.50.250 avec 32 octets de données :
Réponse de 192.168.50.250 : octets=32 temps=2 ms TTL=63
Réponse de 192.168.50.250 : octets=32 temps=1 ms TTL=63

Statistiques Ping pour 192.168.50.250:
    Paquets : envoyés = 2, reçus = 2, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 1ms, Maximum = 2ms, Moyenne = 1ms
Ctrl+C
^C
C:\Users\Admi>ping 192.168.30.45

Envoi d'une requête 'Ping' 192.168.30.45 avec 32 octets de données :
Réponse de 192.168.30.45 : octets=32 temps=3 ms TTL=61
Réponse de 192.168.30.45 : octets=32 temps=2 ms TTL=61
Réponse de 192.168.30.45 : octets=32 temps=2 ms TTL=61
Réponse de 192.168.30.45 : octets=32 temps=2 ms TTL=61

Statistiques Ping pour 192.168.30.45:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 2ms, Maximum = 3ms, Moyenne = 2ms
```

